

ACTIVITY 3 — LEARNER WORKSHEET**Analyze — Find the Root Cause**

TYPE	DURATION	MAPS TO	TOPIC
Hands-on Lab	30 minutes	ANALYZE	Analyze — Find the Cause

Name	
Date	

What you are doing

The team now knows WHERE the delay happens. This activity asks WHY it happens.

What you will produce: A completed 5 Whys chain ending in an actionable cause, a Fishbone diagram with causes sorted into the five categories, and a shortlist of two or three root causes each backed by evidence.

The scenario

At the last team meeting two theories were argued loudly. The owner still wants to buy a second machine. The shift supervisor wants to add a staff member to the morning rota. Both are solutions to a symptom nobody has yet explained: the queue is long, so add capacity. Neither theory has been tested against the observation data.

Most improvement teams jump straight from a symptom to a solution and end up fixing the wrong thing at considerable cost. Your job is to drill past the symptom to a cause the team can actually act on — and to be able to show the evidence for it.

Your steps

1. **State the symptom** — Write the symptom from your Lab 2 baseline at the top of templates/5-whys-template.csv — for example 'customers wait an average of 8.9 minutes against a 5-minute promise'. Start from the measured symptom, not from a theory about the cause.
2. **Drill down with 5 Whys** — Ask 'why does that happen?' down the template, each answer becoming the next question. Stop when you reach a PROCESS the team can act on. Two rules: never stop at a person ('the barista is slow' is not a root cause — ask why), and do not stop early just because an answer sounds plausible.
3. **Set up the Fishbone** — Open templates/fishbone-template.csv and write the problem in the head of the fish. The 5 Whys drills DOWN one chain; the Fishbone spreads WIDE so no whole area of cause is missed. Used together they are stronger than either alone.
4. **Populate the five bones** — Brainstorm possible causes onto Manpower, Method, Machine, Material and Measurement. Then check them against data/cause-evidence-sheet.csv to see which ones the data actually supports — some entries there are pure opinion, and telling them apart is the point.

5. **Read the Pareto** — Open data/delay-reason-pareto-data.csv — the delay reasons from your Lab 2 log, counted and ranked. Read the Cumulative % column and state which few causes account for most of the problem. Remember the Pareto ranks FREQUENCY, which is usually but not always the same as importance.
6. **Separate common from special cause** — Using the last column of the cause-evidence sheet, decide for each cause whether it is common cause (built into the process, happens every day) or special cause (a specific one-off event). Common causes are fixed by changing the process; a team that chases special causes is always firefighting.
7. **Shortlist on evidence** — In templates/root-cause-shortlist-template.csv, shortlist the two or three causes best supported by evidence — not by opinion, and not by who argued loudest. For each, record the evidence and where it came from. Three is the maximum a White Belt team should carry into Improve.

Your working space

Complete every field. You may keep this worksheet and refer to it during the open-book assessment.

Step 1 — State the symptom

Step 2 — Drill down with 5 Whys

Step 3 — Set up the Fishbone

Step 4 — Populate the five bones

Step 5 — Read the Pareto

Step 6 — Separate common from special cause

Step 7 — Shortlist on evidence

Answers to the discussion prompts

1. Compare your 5 Whys chain with another group's. You started from the same symptom — did you reach the same root cause, and what does that tell you?

2. Which causes on your Fishbone have real evidence behind them, and which are assumptions the team simply believes?

3. The Pareto ranks causes by frequency. When might the most frequent cause NOT be the most important one?

4. Why does it matter whether a cause is common cause or special cause?

Reflect & discuss

Did your chain stop at a person at any point? What happened when you asked 'why' again?

Which of the owner's or supervisor's two theories survives contact with the data?

What would it have cost the cafe to act on the loudest opinion instead of the evidence?

Check your work

Every group can show a 5 Whys chain ending in an actionable process cause, a Fishbone with causes sorted into the five categories, and two or three shortlisted causes each backed by a stated observation.